



Ethics of Smart Cities and IoT

Engineering Ethics Module

Introduction to Smart Cities & IoT

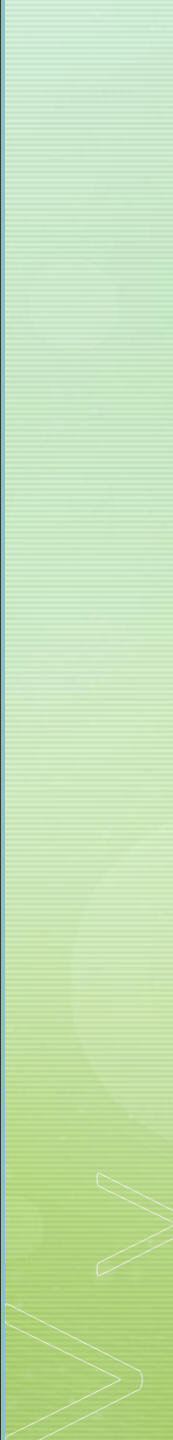
- Smart cities use IoT for data-driven services
- Applications: Traffic, energy, healthcare, surveillance
- Ethical implications are critical for engineers

Why Ethics Matters in Smart Cities

- Privacy concerns with pervasive data collection
- Algorithmic bias and fairness issues
- Security vulnerabilities impacting citizens

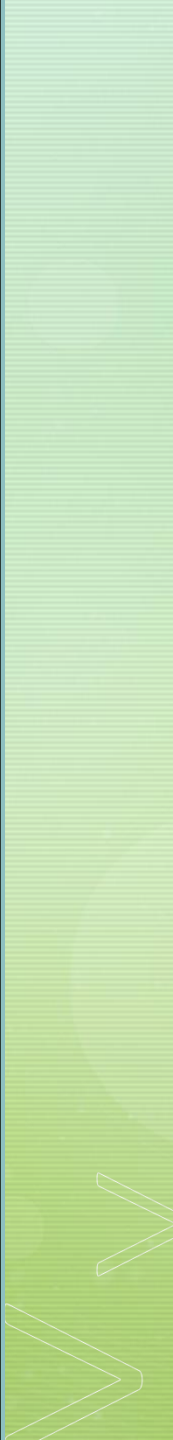


Core Ethical Principles

- Transparency in data collection
 - Accountability of decision-making systems
 - Respect for privacy and consent
 - Non-maleficence and fairness
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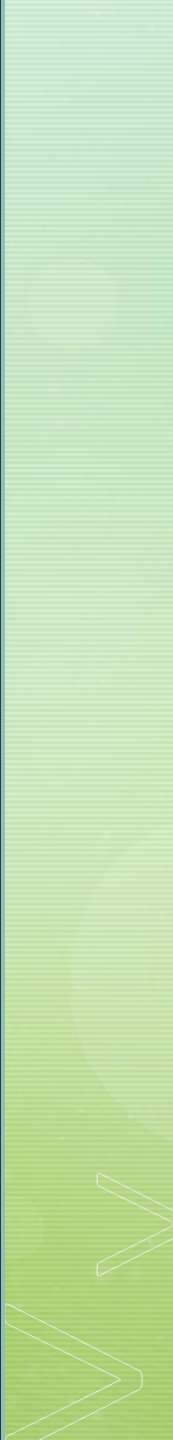


Key IoT Components

- Smart sensors and devices
 - Data analytics platforms
 - AI-driven decision engines
 - Cloud and edge infrastructure
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Ethical Challenges

- Mass surveillance vs public safety
 - Ownership and control of data
 - Algorithmic decision-making opacity
 - Cybersecurity threats
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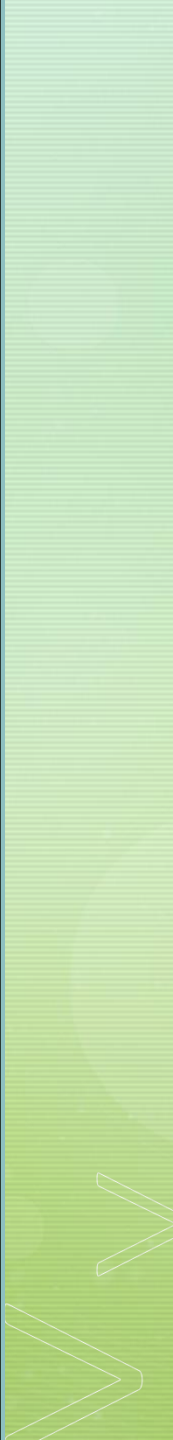


Legal & Policy Frameworks

- GDPR for data privacy
- Local data protection laws
- Smart city governance standards



Case Study 1: Sidewalk Labs (Toronto)

- Google's smart city project faced backlash over data privacy
 - Concerns: Who owns urban data? How is it monetized?
 - Result: Project canceled after ethical and governance concerns
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Case Study 2: Singapore Smart Nation

- Extensive IoT deployment for transport & security
- Debate: Security vs citizen privacy
- Ethical challenge: consent in public surveillance

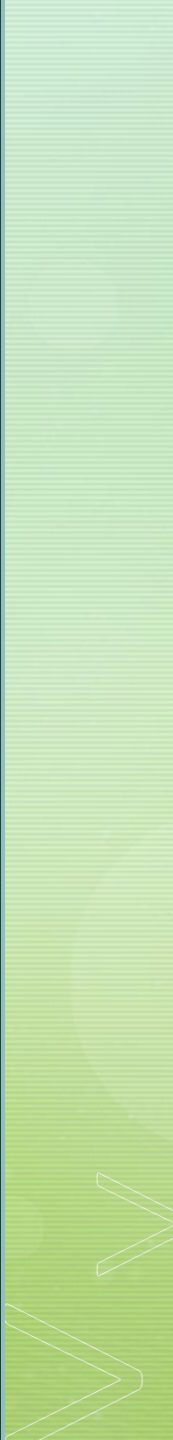


Case Study 3: Indian Smart Cities Mission

- Push for IoT and e-governance
- Challenges: Digital divide and data misuse
- Ethical issues: inclusion and data protection



Debate Prompt

- Should governments install facial recognition in public spaces?
 - Pros: Crime prevention, public safety
 - Cons: Privacy invasion, potential misuse
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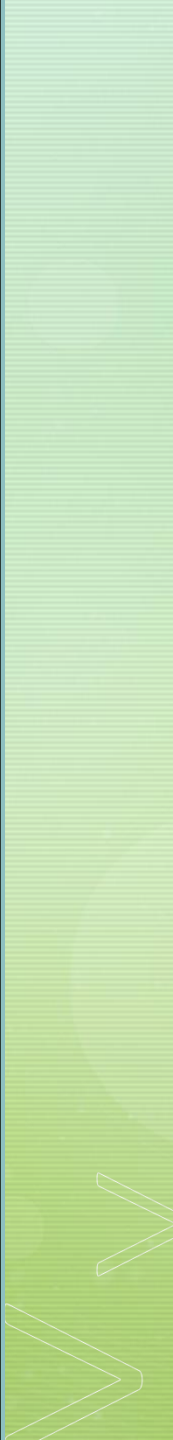
Activity: IoT Risk Analysis

Pick an IoT application (e.g., traffic sensors, smart meters).

Identify 3 potential ethical issues and propose mitigation strategies.

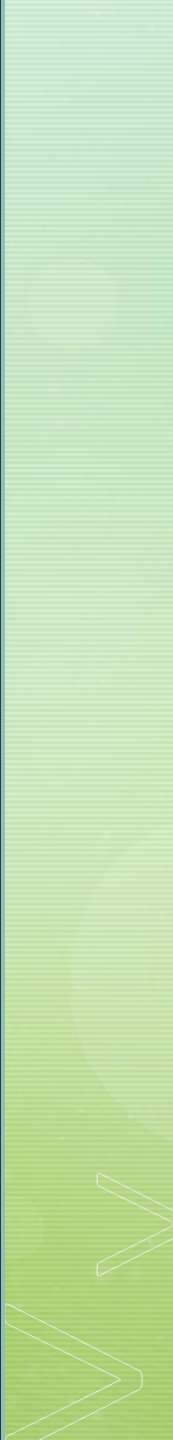


Security & Hacking Risks

- Smart grids hacked → power outages
 - Connected cars hijacked → safety threat
 - Data leaks from health IoT devices
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AI and Algorithmic Bias

- Predictive policing using biased data
 - Resource allocation algorithms discriminating against neighborhoods
 - Ethical duty: Ensure fairness and accountability
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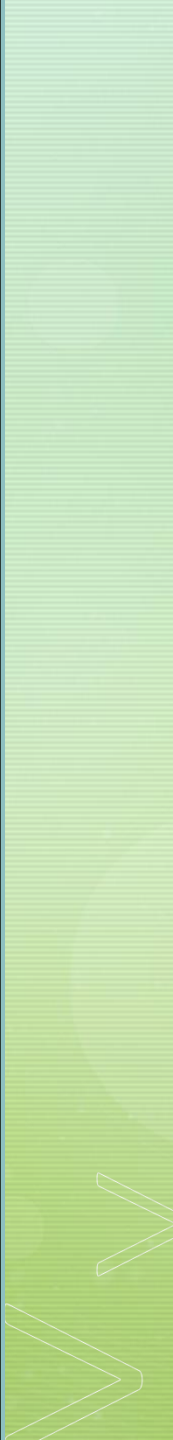


Environmental & Sustainability Ethics

- Energy consumption of IoT devices
- E-waste and recycling concerns
- Green smart city initiatives

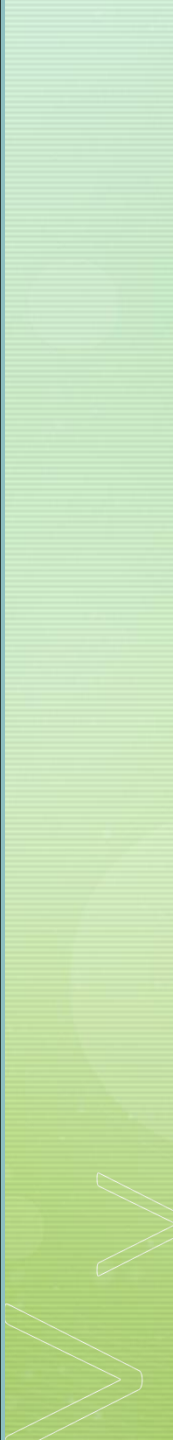


Professional Guidelines

- IEEE IoT ethics standards
 - ACM Code of Ethics
 - Principle: 'Respect privacy, prioritize safety'
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Emerging Tech & Risks

- 5G and increased IoT connectivity
 - Digital twins and predictive models
 - Ethical dilemmas with autonomous systems
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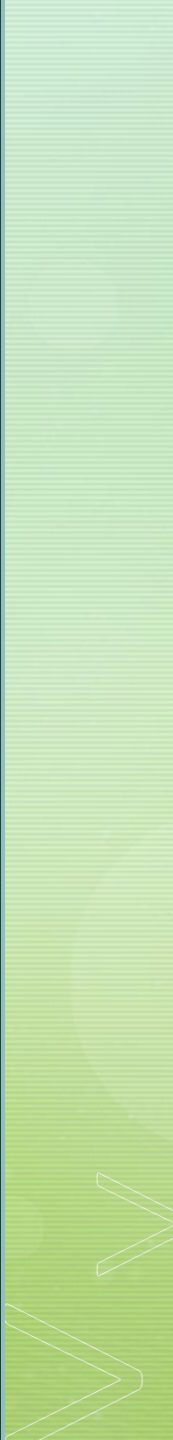
Scenario Analysis

Scenario: A smart traffic system prioritizes VIP routes, delaying ambulances.

Question on fairness and propose ethical fixes.



Best Practices

- Privacy by design
 - Secure IoT architecture
 - Transparency and stakeholder engagement
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Summary & Takeaways

- Smart cities promise efficiency but pose ethical risks
 - Balance innovation with rights and fairness
 - Engineers play a key role in ethical IoT design
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